

Ashutosh Kumar Vishwakarma

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Education

Indian Institute of Information Technology (IIIT), Bhopal

B.Tech in Computer Science and Engineering

Sem VI SPI: 9.85 | Department Rank 1

Aug 2023 – May 2027

CGPA: 8.79/10

Relevant Courses: Data Structures & Algorithms, DBMS, Software Engineering, OOP, Operating System, Computer Networks, Distributed Systems, Cryptography & Information Security

Experience

Software Engineering Intern, QuickIntell | United States

Nov 2025 – March 2026 | Remote

- Implemented JWT-based authentication and session management with strict input validation across RESTful APIs, enforcing secure data contracts between frontend and backend services and closing injection/auth-bypass attack surface.
- Performed data modeling and architected **5 relational database schemas** with Prisma and PostgreSQL, consumed by **4 independently deployed microservices**, incorporating senior engineer code review feedback to align with existing architectural patterns and ensure consistent data contracts.
- Onboarded into an existing production codebase and built an analytics module comprising **11 modules** processing **15+ event data points**, replacing manual data review with real-time visual dashboards used across the team for internal reporting.

Technical Projects

RepoSense – AI-Powered Multi-Agent Codebase Intelligence System

Jul 2026 – Present

Python, LangGraph, MCP, Groq | Multi-Agent AI Systems

[Source](#)

- Building a multi-agent backend system using LangGraph and the Model Context Protocol (MCP); implemented a custom GitHub MCP server exposing repo/file/issue-search tools as callable endpoints, with an LLM agent (Groq-hosted gpt-oss-120b) autonomously selecting and invoking tools based on natural language queries.
- In progress: RAG layer over Qdrant for semantic code search, multi-agent fact-checking pipeline, and automated evaluation harness for scoring agent tool-selection accuracy.

Baskit – Scalable E-commerce Platform

Apr 2025 – May 2025

React, Node.js, Express, MongoDB, Redis, Stripe, Gemini API

[Source](#) | [Live Demo](#)

- Implemented secure payment pipelines using Stripe Webhooks with end-to-end signature/event validation and idempotency handling, preventing replay attacks and duplicate transaction processing under high-concurrency load.
- Enforced strict data consistency via MongoDB transactions across payment and inventory flows under high-concurrency load, validated with Apache JMeter simulating 500+ concurrent transactions; engineered a Redis caching layer cutting database read load by 40%.
- Built a Retrieval-Augmented Generation (RAG) product recommendation pipeline using Google Gemini's Generative AI API and text-embedding-004, executing cosine similarity search on MongoDB Atlas Vector Search to deliver personalized results with sub-50ms latency.

Achievements

- Security:** Discovered and responsibly disclosed a security vulnerability in IIIT Bhopal's official college website, coordinating directly with site administrators on remediation.
- Competitive Programming:** LeetCode Knight (Rating: 1900+); AIR 70 nationally in Naukri Engineers' Ring of Honour; solved 900+ problems (LeetCode, Codeforces, and CodeChef) with emphasis on graphs, dynamic programming, and trees.
- Contest Performance:** Ranked 837/41,000+ (top 3%) in LeetCode Weekly Contest 492 and 887/40,000+ (top 3%) in Weekly Contest 493; Global Ranks 75 (Starters 181), 136 (Starters 176), and 271 (Starters 230) on CodeChef.
- AlgoUniversity Graph Theory Camp:** Selected among top 2.5% of 40,000+ applicants nationwide.
- Inter-IIIT:** Rank 19/1000+ (top 2%), OPTIGO'26 Inter-IIIT Coding Competition, Coding Club CODAME, IIIT Bhopal (2026).

Technical Skills

- Languages:** C++, Python, TypeScript, JavaScript (ES6+), SQL
- Security & Core Competencies:** Authentication (JWT), Input Validation & Sanitization, Secure API Design, OOP and Design, Algorithms, Data Structures, Network Programming (TCP/IP)
- Databases & Cloud:** PostgreSQL, MongoDB, Redis, Docker
- AI/ML:** Generative AI, RAG Pipelines, Vector Search, Embedding Models
- Developer Tools:** Git/GitHub, GitHub Actions (CI/CD), GCC/Clang, Postman, Apache JMeter, WSL, Vercel, AWS
- Frameworks & Libraries:** Next.js 14, React.js, Zustand, Redux Toolkit, Express.js, Node.js, Prisma ORM